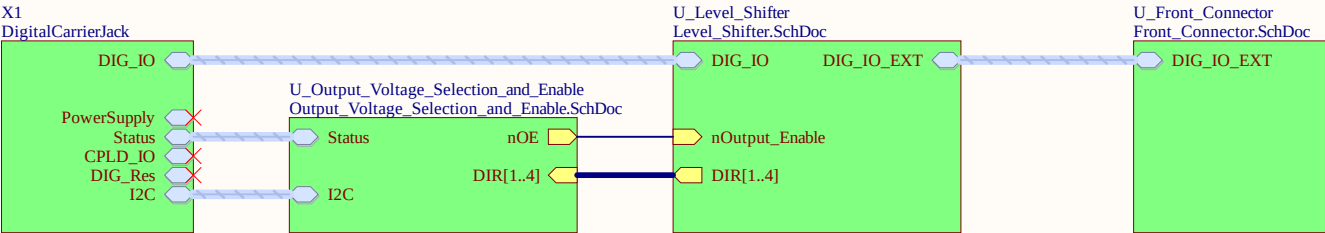


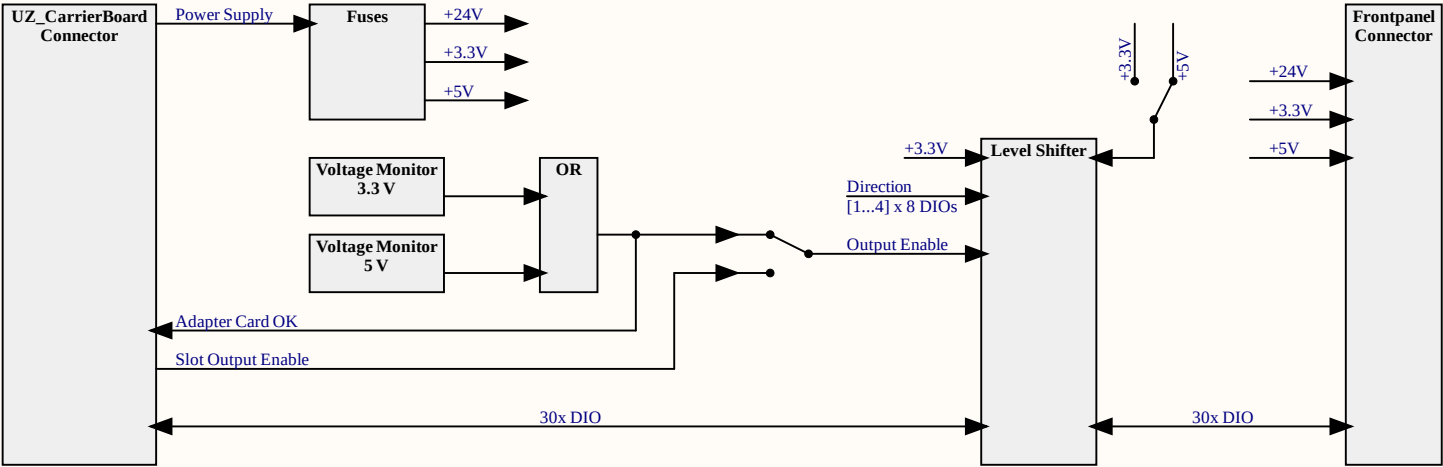
# Top Sheet



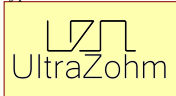
## Block Diagram

Board Features:

- voltage monitor ICs supervise 3.3V and 5V rail
- output enable of level shifters can be controlled either by the supervisors status or via UZ\_CarrierBoard signal
- I<sup>2</sup>C-Port Expander reads output voltage level and signal direction per group of 8 channels
- RGB-LED at front panel visualizes selected voltage and error status



Chose size of UZ logo  
by chosing footprint  
type



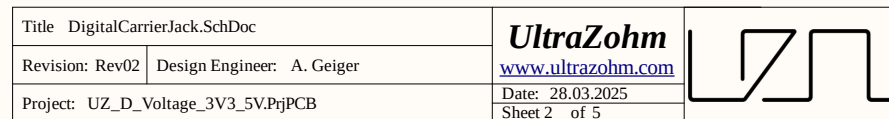
LOGO1

### INFO

Project  
ProjectRevision  
AuthorParam  
ProjectDate

Design Information

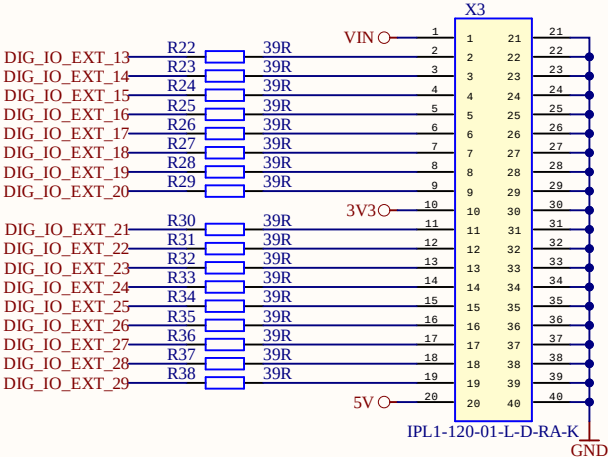
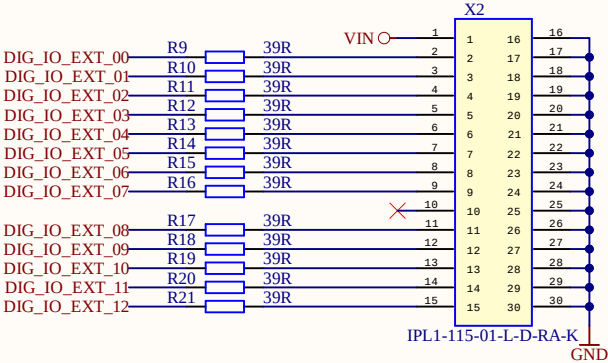
|                                     |                            |                                                                                                                  |                                                                                       |
|-------------------------------------|----------------------------|------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Title TopSheet.SchDoc               |                            | <b>UltraZohm</b><br><a href="http://www.ultrazohm.com">www.ultrazohm.com</a><br>Date: 28.03.2025<br>Sheet 1 of 5 |  |
| Revision: Rev02                     | Design Engineer: A. Geiger |                                                                                                                  |                                                                                       |
| Project: UZ_D_Voltage_3V3_5V.PrjPCB |                            |                                                                                                                  |                                                                                       |



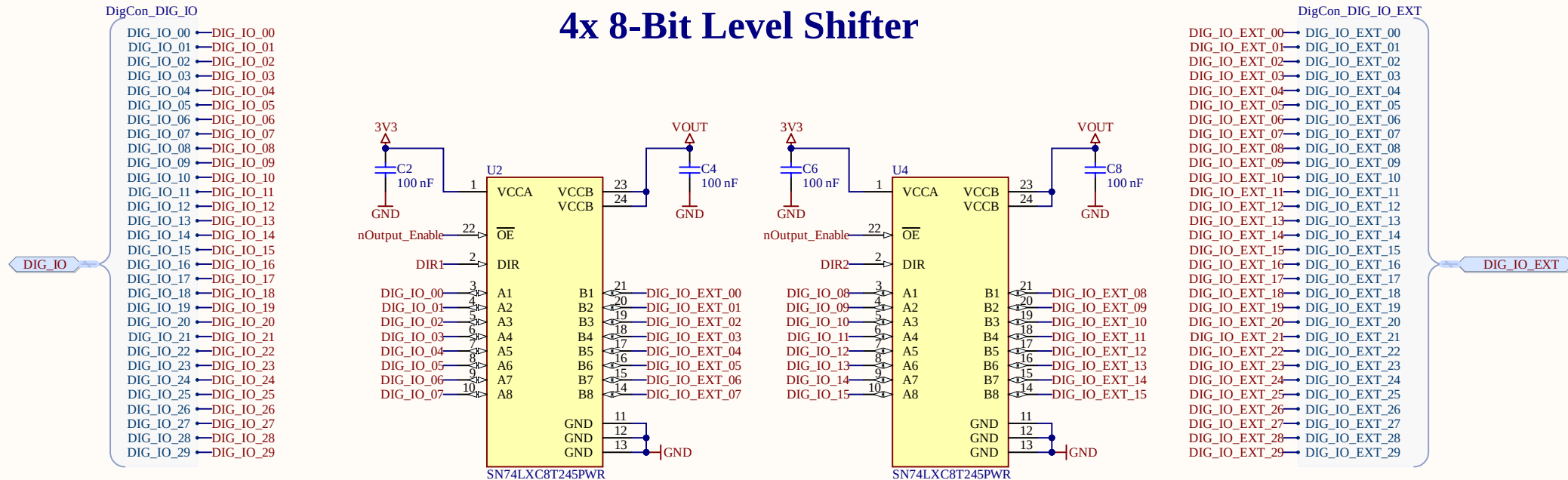
# Front Panel Connector

DigCon\_DIG\_IO\_EXT

- DIG\_IO\_EXT\_00
- DIG\_IO\_EXT\_01
- DIG\_IO\_EXT\_02
- DIG\_IO\_EXT\_03
- DIG\_IO\_EXT\_04
- DIG\_IO\_EXT\_05
- DIG\_IO\_EXT\_06
- DIG\_IO\_EXT\_07
- DIG\_IO\_EXT\_08
- DIG\_IO\_EXT\_09
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- DIG\_IO\_EXT\_11
- DIG\_IO\_EXT\_12
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- DIG\_IO\_EXT\_24
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- DIG\_IO\_EXT\_27
- DIG\_IO\_EXT\_28
- DIG\_IO\_EXT\_29

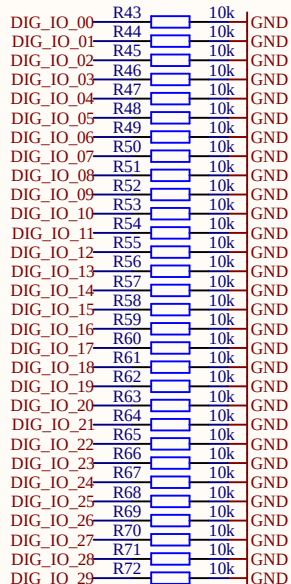


# 4x 8-Bit Level Shifter

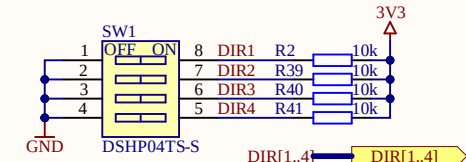


nOutput\_Enable

## Pull-Down Resistors



## Direction per Level Shifter



| Table 8-1. Function Table <sup>(1)</sup> |     |                  |                  |
|------------------------------------------|-----|------------------|------------------|
| CONTROL INPUTS                           |     | Port Status      |                  |
| OE                                       | DIR | A PORT           | B PORT           |
| L                                        | L   | Output (Enabled) | Input (Hi-Z)     |
| L                                        | H   | Input (Hi-Z)     | Output (Enabled) |
| H                                        | X   | Input (Hi-Z)     | Input (Hi-Z)     |

(1) Input circuits of the data I/Os are always active and should be kept at a valid logic level.

| DIP-Switch | DIR  | Signal Direction |
|------------|------|------------------|
| OFF        | HIGH | Output           |
| ON         | LOW  | Input            |

